Supplementary Material to "Regional Favoritism"

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List of countries: The following 126 countries are in our sample (e.g., Table II, column 1, in the paper): Afghanistan, Albania, Algeria, Angola, Argentina, Australia, Austria, Bangladesh, Belarus, Belgium, Benin, Bhutan, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cambodia, Cameroon, Canada, Central African Republic, Chad, Chile, China, Colombia, Costa Rica, Czech Republic, Cote d'Ivoire, Democratic Republic of the Congo, Denmark, East Timor, Ecuador, El Salvador, Eritrea, Ethiopia, Fiji, Finland, France, Gabon, Gambia, Georgia, Germany, Ghana, Greece, Guatemala, Guinea, Guinea-Bissau, Guyana, Haiti, Honduras, India, Indonesia, Iran, Iraq, Italy, Japan, Jordan, Kazakhstan, Kenya, Laos, Latvia, Lebanon, Liberia, Lithuania, Macedonia, Madagascar, Malawi, Malaysia, Mali, Mauritania, Mexico, Mongolia, Morocco, Mozambique, Myanmar, Namibia, Nepal, Netherlands, New Zealand, Nicaragua, Niger, Nigeria, North Korea, Norway, Oman, Pakistan, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Republic of Congo, Russia, Rwanda, Senegal, Serbia, Sierra Leone, Slovakia, Slovenia, Somalia, South Africa, South Korea, Spain, Sri Lanka, Sudan, Sweden, Taiwan, Tajikistan, Tanzania, Thailand, Togo, Tunisia, Uganda, Ukraine, United Kingdom, United States, Uruguay, Venezuela, Vietnam, Yemen, Zambia, Zimbabwe.

The distribution of nighttime light intensity and our dependent variable: Figure S.1 reports the distribution of the average nighttime light intensity in our sample. It shows that the distribution is strongly skewed towards the right, with concentrations at the lower bound (0) and the upper bound (63). Figure S.2 reports the distribution of our main dependent variable, $Light_{ict}$, which corresponds to the logarithm of the average nighttime light intensity plus 0.01. This transformation has also been used by Michalopoulos and Papaioannou (2013a,b) to keep observations with zero average nighttime light intensity and to minimize the problem of outliers.

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Estimates to investigate the role of particularly dark and bright observations: Average nighttime light intensity is equal to zero in 10.27 percent of our observations and equal to 63 in 0.85 percent of our observations. A standard approach in the presence of zeros and top-coding is to use a Tobit-type estimator. Angrist and Pischke (2009, p. 102) however argue that Tobit-type estimators only make sense if the data is truly censored. In our case, the zeros are not censored. Zero reported nighttime light does not represent some negative latent nighttime light, but most likely positive nighttime light that was not detected by the satellites, or wrongly identified as ephemeral light or background noise and therefore set to zero by the NOAA. The observations with average nighttime light intensity of 63 are truly censored, but they constitute less than one percent of our sample. We are therefore doubtful whether a Tobit-type estimator is appropriate. We nevertheless use Honore's (1992) panel Tobit estimator to check the robustness of our results. This estimator addresses the incidental parameter problem in the presence of one-way fixed effects. Table S.1 presents the two robustness exercises mentioned in footnote 14 in the paper. We use the logarithm of average nighttime light intensity plus 1 (rather than 0.01) as dependent variable, because the panel Tobit estimator works for a lower bound of the dependent variable of 0, but not for a negative lower bound. Column 1 presents results from using the panel Tobit estimator without any time dummy variables, and column 2 when including year dummy variables. The coefficient of interest is positive and statistically significant in both columns. For comparison purposes, columns 3 and 4 present the results from standard least squares with fixed effects without and with year dummy variables. The results are very similar. Hence, our results do not seem to be driven by the reliance on standard least squares with fixed effects.[2]

As an additional robustness exercise, we test the sensitivity of our results to the exclusion of the brightest and darkest quartiles in our sample. Columns 5 and 6 show the results from our main specification (Table II, column 1, in the paper) when dropping the quartile of observations from the regions with the highest average nighttime light intensity

[2]The panel Tobit estimator does not converge if we include the full set of more than 2000 country-year dummy variables. However, it is not clear whether including all these dummy variables would be recommended given that Honore (1992) designed his panel Tobit estimator to deal with the incidental parameter problem in the case of one-way fixed effects, but not in the case of two-way fixed effects.

(averaged over the sample period), and the quartile of observations from the regions with the lowest average nighttime light intensity (averaged over the sample period). In both cases, the coefficient of interest remains statistically significant and is at least as high as in our main specification. Hence, our results are not driven by regions in which average nighttime light intensity is particularly low or high.

Column 7 adds an interaction term between $Leader_{ict-1}$ and the regional population density to our main specification, as observations are more likely to have no recorded nighttime light in sparsely populated regions, and very intense nighttime light in densely populated regions. We find a positive and statistically significant effect of being the leader region on nighttime light intensity even in barely populated regions, and this effect becomes larger as population density increases.

Estimates with nighttime light intensity per capita as dependant variable: Tables S.2–S.7 present the equivalent of Tables II–VII in the paper, but with the logarithm of nighttime light intensity per capita as dependent variable, and controlling for the logarithm of regional population. Doing so is associated with some caveats in all cases (see footnote 15 in the paper), and is not meaningful for the regressions in Table IV, columns 1 and 4-7, where the units of observation are not based on administrative regions.

Comparing Tables S.2–S.7 to Tables II–VII shows that almost all specifications lead to qualitatively similar results when using the logarithm of nighttime light intensity per capita instead of $Light_{ict}$ as dependent variable. Exceptions are column 5 of Tables III and S.3, and column 4 of Tables VII and 7.

IV estimates for the effects of foreign aid inflows and oil rents: Table S.8 presents the same specifications as Table VII in the paper, but uses an instrumental variables approach to address the potential endogeneity of foreign aid inflows and oil rents. As the region fixed effects control for all time-invariant determinants of foreign aid inflows and oil rents, we use instrumental variables that capture exogenous variation over time. Using bilateral foreign aid flows from 22 donors to all recipient countries, Harding and Venables (2010) compute constructed aid inflows of recipient country c as

$\sum_{d=1}^{22} (s_{cd} a_{dt})$, where s_{cd} is the average annual share of bilateral aid that donor d gave to this recipient country over the period from 1960 to 2008, and a_{dt} is total bilateral aid from donor d to all recipient countries in year t.[3] We compute constructed oil rents as the time-averaged oil production of country c multiplied by the difference between the world market price of oil in year t and the time-averaged extraction costs of country c. Our instrumental variables, Aid_{ct}^{IV} and Oil_{ct}^{IV}, are the logarithms of these constructed aid inflows and oil rents. These instrumental variables exploit the arguably exogenous variation in the donors' aid budgets and the world market price of oil over time.[4] All first-stage regressions are presented in Table S.9.

Columns 1 and 2 of Table S.8 show that the IV estimates of the coefficients on $Leader_{ict-1} \times Aid_{ct-1}$ and $Leader_{ict-1} \times Oil_{ct-1}$ are very similar to the corresponding OLS estimates in the paper. Hence, these estimates also suggest that foreign aid does, on average, promote rent seeking and regional favoritism, while oil does not. The IV estimates in columns 3 and 4 of Table S.8 also show the same pattern as the corresponding OLS estimates in the paper, i.e., a tendency of foreign aid inflows and oil rents do promote rent seeking and regional favoritism in countries with weak political institutions, but not in countries with strong political institutions. While the OLS estimates suggest that this pattern is stronger for foreign aid inflows, the IV estimates here suggest that it is stronger for oil rents.

To illustrate this pattern, we look at the effects of oil rents in Norway and Oman. These countries have the highest oil rents per capita in our sample ($Oil_{NOR2008} = 9.46$ and $Oil_{OMN2008} = 9.07$), but remarkably different political institutions ($Polity_{NOR2008} = 1.0$ and $Polity_{OMN2008} = 0.1$). The coefficient estimates in column 4 of Table S.8 suggest that oil rents fuel regional favoritism in Oman, but not in Norway. These predictions match well with these countries' experiences. Oman, on the one hand, has long been ruled by Sultan Qaboos bin Said. He and many of his ministers come from Salalah in Dhofar province. There have been accusations that significant parts of the oil revenues are channeled towards Dhofar (rather than more populated provinces), or embezzled by

[3]We thank Torfinn Harding for sharing their data.

[4]Saudi Arabia, which is the largest oil producer, is not in our sample.

corrupt ministers.[5] Norway, on the other hand, is a well known success story in handling its oil revenues, and its well-established democratic institutions are generally seen as one of the main reasons (e.g., Mehlum et al., 2012).[6]

The effects of natural disasters: Table S.10 investigates the effect of negative shocks in the form of natural disasters on the extent of regional favoritism. Natural disasters might put a temporary constraint on the public funds available for regional favoritism because of a decrease in public revenues and a reallocation of public funds required for recovery. We use data on the occurrence of natural disasters from EM-DAT by the Center for Research on the Epidemiology of Disasters (CRED) in Brussels. EM-DAT is the most comprehensive collection of disaster events and contains around 12,000 reports of different disasters, such as floods, storms, earthquakes, volcanic eruptions, landslides as well as man-made disasters. A natural disaster has to fulfil at least one of the following criteria in order to be included in the database: 10 or more people reported killed, 100 or more people reported affected, declaration of a state of emergency, or call for international assistance. We exclude man-made disasters. From this dataset, we construct two different disaster variables. First, $Disaster_{ct}$ is a dummy variable that is equal to one if a natural disaster hit country c in year t, and to zero otherwise. Although, the EM-DAT's definition of a disaster already excludes small events, there are some countries that suffer from a disaster almost every year. We therefore construct a second dummy variable, $DisasterLarge_{ct}$ that is equal to one if and only if the disaster has affected more than 1,000 people. We then build the interaction terms $Leader_{ict} \times Disaster_{ct}$ and $Leader_{ict} \times DisasterLarge_{ct}$.

We would like to interpret the coefficient on these interaction terms as the effects of a negative shock to public funds on regional favoritism. However, political leaders tend to come from relatively populous regions, and cities are often located in areas that are

[5]See, e.g., The Economist, "Where's our sultan?", August 7, 1997; and The Economist, "The sultanate suddenly stirs", March 3, 2011.

[6]Simple correlations between oil rents and subsequent nighttime light intensity are also consistent with these predictions: The correlation between Oil_{ct-1} and $Light_{ict}$ is 0.90 for Salalah (but high in some other Omani regions as well). Unlike Oman, Norway had several political leaders during our sample period. The longest serving political leader was Kjell Magne Bondevik from Molde, who was in office from 1997 to 2005. The correlation between Oil_{ct-1} and $Light_{ict}$ is -0.05 for Molde during these years (and close to zero in most other Norwegian regions as well).

more exposed to natural hazards (e.g., at the coast or inland waterways). Therefore, a negative coefficient on these interaction terms could also reflect that leader regions are more likely to be among the affected regions. To account for this, we construct a third variable that controls for region i's exposure to natural disasters. This GIS-based data is taken from Dilley et al. (2005) that contains georeferenced data about subnational disaster mortality risk. The variable basically combines a region's exposure to multiple hazards with historical information (1981-2000) on the number of people killed in disasters in the exposed area to construct a proxy for the differences in mortality of disasters between areas. A mortality rate was calculated by dividing the number of historical disaster fatalities in the area by the total population living in the exposed area. This ratio was then converted into an index ranging from 1-10 that classifies the global multihazard mortality into deciles. For each region, i, we calculate the average mortality risk, $DisMortality_{ic}$, and interact it with $Disaster_{ct}$ as well as $DisasterLarge_{ct}$.

The results are presented in Table S.10. Column 1 only includes the leader variable and its interaction with the general disaster dummy. The coefficient estimates imply that being the leader region increases average nighttime light intensity by 6.7 percent during non-disaster years, and by 2.4 percent during disaster years. Controlling for the disaster mortality risk in each region does not change the results qualitatively or quantitatively. The estimates using large disasters alone in columns 3 and 4 indicate a somewhat smaller constraining effect of disasters, which is no longer statistically significant at conventional levels. Overall, we take the results presented in Table S.10 as tentative evidence that natural disasters seem to correspond to a negative shock to the public funds available to the political leaders and thereby tend to reduce the extent of preferential treatment of the political leader's birth region during disaster years.

The effects of the political leaders' characteristics: Table S. 11 analyzes whether and how the political leaders' characteristics impact on regional favoritism. We obtained information about the leaders' current Age_{ct} directly from Archigos. Data on the leader's education and prior occupation was obtained from Besley and Reynal-Querol (2011).[7] We

[7]We thank Marta Reynal-Querol for sharing their data.

use the following variables: $Graduate_{ct}$, which is the key variable in Besley and Reynal-Querol's study, is a dummy variable that is equal to one if the current political leader has a graduate degree. $Military_{ct}$ is a dummy variable that is equal to one if the leader's prior occupation was with the military; $ProfScien_{ct}$ is a dummy variable that is equal to one if the leader was a professor or scientist; and $Lawyer_{ct}$ is a dummy variable that is equal to one if the leader was a lawyer. We then use interaction terms between $Leader_{ict-1}$ and each of these variables capturing leaders' characteristics. As Archigos and the data from Besley and Reynal-Querol only contain information about political leaders up to 2004, we restricted our sample to the period 1992-2005 for this exercise (and we drop country-years for which characteristics of leaders are missing). The estimation results for specifications that include $Leader_{ict-1}$ and the different interaction terms are presented in Table S.11. None of these interaction terms, except $Leader_{ict-1} \times Military_{ct-1}$ in column 4, is statistically significant. Therefore, we do not find much evidence for a systematic impact of the leaders' characteristics on the extent of regional favoritism.

Nighttime light in Gbadolite, Hambantota and comparable towns: Figures I and II in the paper show nighttime light intensity in Gbadolite and Hambantota before and after changes in the leadership in Zaire (DRC) and Sri Lanka, respectively. The changes visible in these figures can reflect location-specific or broader changes, including changes in the satellites' sensor settings (see footnote 9 in the paper). Therefore, Figures S.3 and S.4 complement these figures with the development of nighttime light intensity in nearby towns with similar luminosity. These towns are Kinsangani in Figure S.3, and Pottuvil in Figure S.4. These new figures suggest that a considerable part, but not all of the changes in Gbadolite and Hambantota are location-specific and, therefore, the likely consequence of changes in leadership.

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Tables and Figures

Table S.1: Estimates to investigate the role of particularly dark and bright observations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	$Light1_{ict}$	$Light1_{ict}$	$Light1_{ict}$	$Light1_{ict}$	$Light_{ict}$	$Light_{ict}$	$Light_{ict}$
$Leader_{ict-1}$	0.023*	0.024**	0.021***	0.021***	0.051**	0.038**	0.029**
	(0.013)	(0.012)	(0.007)	(0.007)	(0.024)	(0.015)	(0.015)
$Leader_{ict-1} \times PopDen_{ict-1}$							0.000***
							(0.000)
Number of regions	38,427	38,427	38,427	38,427	28,820	28,801	37,475
Observations	690,495	690,495	690,495	690,495	517,874	517,879	673,382
R-squared			0.000	0.208	0.340	0.374	0.327
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	No	No	No
Country-year FE	No	No	No	No	Yes	Yes	Yes
Panel tobit estimator	Yes	Yes	No	No	No	No	No
Excluded observations	None	None	None	None	Brightest quartile	Darkest quartile	None

Notes: Regressions using Honore's (1992) panel Tobit estimator in columns 1-2, and standard fixed effect regressions in columns 3-7, all using annual data for subnational regions between 1992 and 2009. The quartile of observations from the regions with the highest average nighttime light intensity (averaged over the sample period) is excluded in column 5, and the quartile of observations from the regions with the lowest average nighttime light intensity (averaged over the sample period) is excluded in column 6. $Light1_{ict}$ is the logarithm of average nighttime light intensity plus 1. $Light_{ict}$ is the logarithm of average nighttime light intensity plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. $PopDen_{ict}$ is regional population density. Standard errors are adjusted for leader clustering in columns 3-7. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.2: Main Results and Robustness Tests using $Lightpc_{ict}$

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc0_{ict}$
$Leader_{ict-1}$	0.062**			0.035*	0.166***	0.031**
	(0.024)			(0.019)	(0.023)	(0.013)
$Leader_{ict}$		0.063***				
		(0.024)				
$Leader_{ict-2}$			0.060***			
			(0.022)			
$Lightpc_{ict-1}$				0.349***	0.849***	
				(0.021)	(0.012)	
Pop_{ict}	-0.958***	-0.958***	-0.955***	-0.547***	-0.016***	-0.885***
	(0.066)	(0.066)	(0.066)	(0.050)	(0.006)	(0.049)
Number of regions	37,475	37,475	37,475	37,475	37,475	35,669
Observations	673,382	673,382	672,765	636,194	636,194	603,495
R-squared	0.197	0.197	0.197	0.285	0.915	0.338
Region FE	Yes	Yes	Yes	Yes	No	Yes
Country-year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Fixed effect regressions (except for column 5, which is standard OLS) using annual data for subnational regions between 1992 and 2009. $Lightpc_{ict}$ is the logarithm of nighttime light intensity per capita plus 0.01. $Lightpc0_{ict}$ is the logarithm of nighttime light intensity per capita (without adding a constant). $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. Pop_{ict} is the logarithm of regional population. Appendix A in the paper contains more information and sources. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.3: The Dynamics of Regional Favoritism using $Lightpc_{ict}$

	(1)	(2)	(3)	(4)	(5)
Dependent variable	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$
$Leader_{ict-1}$	0.063**	0.065*	0.041	0.010	0.012
	(0.027)	(0.036)	(0.040)	(0.046)	(0.047)
$Leader_{ict-1} \times Experience_{ct-1}$			0.005*		0.001
			(0.003)		(0.005)
$Leader_{ict-1} \times TotalTenure_{ct-1}$				0.008**	0.006
				(0.003)	(0.006)
$Future1_{ict-1}$	-0.021				
	(0.047)				
$Future3_{ict-1}$		0.007	0.026	0.026	0.026
		(0.044)	(0.059)	(0.059)	(0.059)
$Pretrend_{ict-1}$			-0.019	-0.019	-0.019
			(0.031)	(0.031)	(0.031)
$Past1_{ict-1}$	0.026				
	(0.043)				
$Past3_{ict-1}$		0.003	0.012	0.014	0.014
		(0.035)	(0.046)	(0.046)	(0.046)
$Posttrend_{ict-1}$			-0.006	-0.006	-0.006
			(0.023)	(0.023)	(0.023)
Pop_{ict}	-0.958***	-0.958***	-0.958***	-0.958***	-0.958***
	(0.066)	(0.066)	(0.066)	(0.066)	(0.066)
Number of regions	37,475	37,475	37,475	37,475	37,475
Observations	673,382	673,382	673,382	673,382	673,382
R-squared	0.197	0.197	0.197	0.197	0.197
Region FE	Yes	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes	Yes

Notes: Fixed effect regressions using annual data for subnational regions between 1992 and 2009. $Lightpc_{ict}$ is the logarithm of nighttime light intensity per capita plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. $Experience_{ct}$ is the number of years the political leader has been in power until year t. $TotalTenure_{ct}$ is the total number of years the political leader has been in power up to 2010. $Future1_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in $t+1$, but not in t; and 0 otherwise. $Future3_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in $t+1$, $t+2$ or $t+3$, but not in t; and 0 otherwise. $Pretrend_{ict}$ is a time trend for the years in which $Future3_{ict} = 1$. $Past1_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in $t-1$, but not in t; and 0 otherwise. $Past3_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in $t-1$, $t-2$ or $t-3$, but not in t; and 0 otherwise. $Posttrend_{ict}$ is a time trend for the years in which $Past3_{ict} = 1$. Pop_{ict} is the logarithm of regional population. Appendix A in the paper contains more information and sources of all variables used. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.4: The Geographic Extent of Regional Favoritism using $Lightpc_{ict}$

	(2)	(3)
Dependent variable	$Lightpc_{ict}$	$Lightpc_{ict}$
$Leader_{ict-1}$	0.055***	0.035*
	(0.019)	(0.020)
Pop_{ict}	-1.111***	-1.026***
	(0.079)	(0.076)
Number of regions	2,239	2,218
Observations	40,103	39,713
R-squared	0.410	0.408
Region FE	Yes	Yes
Country-year FE	Yes	Yes
Units of observation	SN1 regions	SN1 regions without SN2 leader regions

Notes: Fixed effect regressions using annual data for subnational regions between 1992 and 2009. The units of observations differ across columns and are indicated in the last row (see Section IV in the paper for details). $Lightpc_{ict}$ is the logarithm of nighttime light intensity per capita plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. Pop_{ict} is the logarithm of regional population. In column 3, pixels from SN2 regions that ever were a leader region during our sample period are omitted when constructing $Lightpc_{ict}$ and Pop_{ict} for the corresponding SN1 regions. Appendix A in the paper contains more information and sources of all variables used. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.5: Determinants of Regional Favoritism using $Lightpc_{ict}$

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$
$Leader_{ict-1}$	0.399***	0.183***	0.272*	0.027	0.014	0.192
	(0.104)	(0.068)	(0.155)	(0.030)	(0.017)	(0.197)
$Leader_{ict-1} \times Polity_{ct-1}$	-0.442***					-0.272***
	(0.116)					(0.102)
$Leader_{ict-1} \times Schooling_{ct-1}$		-0.019**				-0.018*
		(0.008)				(0.010)
$Leader_{ict-1} \times NationalGDP_{ct-1}$			-0.025			0.026
			(0.017)			(0.027)
$Leader_{ict-1} \times Language_c$				0.100		-0.025
				(0.076)		(0.062)
$Leader_{ict-1} \times FamilyTies_c$					0.068	0.040
					(0.045)	(0.045)
Pop_{ict}	-0.958***	-0.965***	-0.958***	-0.940***	-0.937***	-0.912***
	(0.067)	(0.068)	(0.066)	(0.065)	(0.067)	(0.067)
Number of regions	37,475	35,102	37,229	36,875	29,788	28,320
Observations	667,314	631,494	666,616	662,582	535,842	505,861
R-squared	0.197	0.201	0.198	0.196	0.178	0.176
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Fixed effect regressions using annual data for subnational regions between 1992 and 2009. $Lightpc_{ict}$ is the logarithm of nighttime light intensity per capita plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. $Polity_{ct}$ is the Polity2 score, rescaled so that it ranges from 0 to 1, with higher values implying stronger political institutions. $Schooling_{ct}$ is the average years of schooling attained. $NationalGDP_{ct}$ is the logarithm of GDP per capita. $Language_c$ is the index of linguistic fractionalization. $FamilyTies_c$ is a measure of the strength of family ties. Pop_{ict} is the logarithm of regional population. Appendix A in the paper contains more information and sources of all variables used. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.6: Regional Favoritism across Continents using $Lightpc_{ict}$

Dependent variable	(1) $Lightpc_{ict}$	(2) $Lightpc_{ict}$	(3) $Lightpc_{ict}$
$Leader_{ict-1} \times Africa_c$	0.112**	0.361***	0.282
	(0.050)	(0.091)	(0.231)
$Leader_{ict-1} \times Americas_c$	0.001	0.370***	0.312
	(0.042)	(0.122)	(0.277)
$Leader_{ict-1} \times Asia_c$	0.166**	0.447***	0.242
	(0.070)	(0.133)	(0.217)
$Leader_{ict-1} \times Europe_c$	-0.015	0.380***	0.291
	(0.011)	(0.122)	(0.238)
$Leader_{ict-1} \times Oceania_c$	-0.056	0.272	0.402*
	(0.163)	(0.208)	(0.240)
$Leader_{ict-1} \times Polity_{ct-1}$		-0.423***	-0.294***
		(0.129)	(0.101)
$Leader_{ict-1} \times Schooling_{ct-1}$			-0.022**
			(0.010)
$Leader_{ict-1} \times NationalGDP_{ct-1}$			0.021
			(0.031)
$Leader_{ict-1} \times Language_c$			-0.008
			(0.056)
$Leader_{ict-1} \times FamilyTies_c$			0.017
			(0.047)
Pop_{ict}	-0.958***	-0.958***	-0.912***
	(0.066)	(0.067)	(0.067)
Number of regions	37,475	37,475	28,320
Observations	673,382	667,314	505,861
R-squared	0.197	0.197	0.176
Region FE	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes

Notes: Fixed effect regressions using annual data for subnational regions between 1992 and 2009. $Lightpc_{ict}$ is the logarithm of nighttime light intensity per capita plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. $Africa_c$, $Americas_c$, $Asia_c$, $Europe_c$, and $Oceania_c$ are dummy variables for the respective continents. $Polity_{ct}$ is the Polity2 score, rescaled so that it ranges from 0 to 1, with higher values implying stronger political institutions. $Schooling_{ct}$ is the average years of schooling attained. $NationalGDP_{ct}$ is the logarithm of GDP per capita. $Language_c$ is the index of linguistic fractionalization. $FamilyTies_c$ is a measure of the strength of family ties. Pop_{ict} is the logarithm of regional population. Appendix A in the paper contains more information and sources of all variables used. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.7: Aid, Oil, and Regional Favoritism using $Lightpc_{ict}$

	(1)	(2)	(3)	(4)
Dependent variable	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$	$Lightpc_{ict}$
$Leader_{ict-1}$	-0.036*	0.046	0.016	0.320**
	(0.022)	(0.030)	(0.129)	(0.144)
$Leader_{ict-1} \times Aid_{ct-1}$	0.013***		0.041**	
	(0.004)		(0.018)	
$Leader_{ict-1} \times Oil_{ct-1}$		-0.002		-0.007
		(0.003)		(0.014)
$Leader_{ict-1} \times Polity_{ct-1}$			-0.061	-0.336**
			(0.131)	(0.168)
$Leader_{ict-1} \times Aid_{ct-1} \times Polity_{ct-1}$			-0.041*	
			(0.020)	
$Leader_{ict-1} \times Oil_{ct-1} \times Polity_{ct-1}$				0.006
				(0.017)
Pop_{ict}	-0.958***	-0.978***	-0.958***	-0.977***
	(0.066)	(0.070)	(0.067)	(0.071)
Number of regions	37,475	37,229	37,475	36,901
Observations	673,382	629,028	667,314	625,256
R-squared	0.197	0.204	0.197	0.204
Region FE	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes

Notes: Fixed effect regressions using annual data for subnational regions between 1992 and 2009. $Lightpc_{ict}$ is the logarithm of nighttime light intensity per capita plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in t; and 0 otherwise. $Polity_{ct}$ is the Polity2 score, rescaled so that it ranges from 0 to 1, with higher values implying stronger political institutions. Aid_{ct} is the logarithm of net overseas development assistance per capita plus 1 (see Appendix A for details). Oil_{ct} is the logarithm of oil rents per capita plus 1. Pop_{ict} is the logarithm of regional population. Appendix A in the paper contains more information and sources of all variables used. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.8: IV Estimates for Aid, Oil, and Regional Favoritism

	(1)	(2)	(3)	(4)
Dependent variable	$Light_{ict}$	$Light_{ict}$	$Light_{ict}$	$Light_{ict}$
$Leader_{ict-1}$	-0.029	0.027	-0.183	0.085
	(0.022)	(0.022)	(0.276)	(0.078)
$Leader_{ict-1} \times Aid_{ct-1}$	0.008**		0.045	
	(0.004)		(0.033)	
$Leader_{ict-1} \times Oil_{ct-1}$		-0.001		0.014
		(0.003)		(0.008)
$Leader_{ict-1} \times Polity_{ct-1}$			0.183	-0.063
			(0.281)	(0.090)
$Leader_{ict-1} \times Aid_{ct-1} \times Polity_{ct-1}$			-0.051	
			(0.034)	
$Leader_{ict-1} \times Oil_{ct-1} \times Polity_{ct-1}$				-0.020**
				(0.010)
Observations	645,319	645,396	642,284	641,410
Region FE	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes
F-statistic	460,048	5,971,600	47,186	2,322,649

Notes: 2SLS regressions using annual data for subnational regions between 1992 and 2009. $Light_{ict}$ is the logarithm of average regional nighttime light plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in t; and 0 otherwise. $Polity_{ct}$ is the Polity2 score, rescaled so that it ranges from 0 to 1, with higher values implying stronger political institutions. Aid_{ct} is the logarithm of net overseas development assistance per capita plus 1 (see Appendix A for details). Oil_{ct} is the logarithm of oil rents per capita plus 1. Appendix A in the paper contains more information and sources of these variables. In columns 1 and 3, $Leader_{ict} \times Aid_{ct}$ (and $Leader_{ict} \times Aid_{ct} \times Polity_{ct}$) are instrumented by $Leader_{ict} \times Aid_{ct}^{IV}$ (and $Leader_{ict} \times Aid_{ct}^{IV} \times Polity_{ct}$), where Aid_{ct}^{IV} is the logarithm of constructed aid inflows per capita plus 1. In columns 2 and 4, $Leader_{ict} \times Oil_{ct}$ (and $Leader_{ict} \times Oil_{ct} \times Polity_{ct}$) are instrumented by $Leader_{ict} \times Oil_{ct}^{IV}$ (and $Leader_{ict} \times Oil_{ct}^{IV} \times Polity_{ct}$), where Oil_{ct}^{IV} is the logarithm of constructed oil rents per capita plus 1. F-statistic is the minimum eigenvalue of a matrix analogue of the first-stage test F-statistic as proposed by Stock et al. (2002). It exceeds the critical value in all columns, suggesting that the instrumental variables are not weak. Standard errors are adjusted for leader clustering. To allow for leader clustering, the 2SLS regressions are run after demeaning all variables at both the regional and the country-year level. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.9: First-stage Estimates for Table S.8

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	$Leader_{ict-1} \times$ Aid_{ct-1}	$Leader_{ict-1} \times$ Oil_{ct-1}	$Leader_{ict-1} \times$ Aid_{ct-1}	$Leader_{ict-1} \times$ $Aid_{ct-1} \times Polity_{ct-1}$	$Leader_{ict-1} \times$ Oil_{ct-1}	$Leader_{ict-1} \times$ $Oil_{ct-1} \times Polity_{ct-1}$
$Leader_{ict-1}$	2.371***	-0.291***	7.987***	2.234***	-0.004	0.128*
	(0.177)	(0.049)	(0.749)	(0.325)	(0.114)	(0.065)
$Leader_{ict-1} \times Polity_{ct-1}$			-6.355***	-0.610*	-0.389**	-0.505***
			(0.787)	(0.369)	(0.157)	(0.118)
$Leader_{ict-1} \times Aid^{IV}_{ct-1}$	2.396***		0.595***	-0.737***		
	(0.061)		(0.219)	(0.104)		
$Leader_{ict-1} \times Aid^{IV}_{ct-1} \times Polity_{ct-1}$			2.068***	3.412***		
			(0.245)	(0.139)		
$Leader_{ict-1} \times Oil^{IV}_{ct-1}$		1.009***			0.991***	-0.017***
		(0.005)			(0.010)	(0.006)
$Leader_{ict-1} \times Oil^{IV}_{ct-1} \times Polity_{ct-1}$					0.025**	1.033***
					(0.012)	(0.009)
Observations	645,319	645,396	642,284	642,284	641,410	641,410
Shea's Partial R-squared	0.416	0.903	0.134	0.163	0.880	0.886
First-stage F-statistic	838	35,678	473	490	18,133	13,531
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Column 1 reports first-stage regressions for Table S.8, column 1; column 2 for Table S.8, column 2; columns 3 and 4 for Table S.8, column 3; and columns 5 and 6 for Table S.8, column 4. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in t; and 0 otherwise. $Polity_{ct}$ is the Polity2 score, rescaled so that it ranges from 0 to 1, with higher values implying stronger political institutions. Aid_{ct} is the logarithm of net overseas development assistance per capita plus 1 (see Appendix A for details). Oil_{ct} is the logarithm of oil rents per capita plus 1. Aid^{IV}_{ct} is the logarithm of constructed aid inflows per capita plus 1. Oil^{IV}_{ct} is the logarithm of constructed oil rents per capita plus 1. First-stage F-statistic is the F-statistic for the joint significance of the coefficients of the instrumental variables. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.10: Natural Disasters and Regional Favoritism

	(1)	(2)	(3)	(4)
Dependent variable	$Light_{ict}$	$Light_{ict}$	$Light_{ict}$	$Light_{ict}$
$Leader_{ict-1}$	0.065***	0.065***	0.052***	0.052***
	(0.021)	(0.021)	(0.018)	(0.018)
$Leader_{ict-1} \times Disaster_{ct-1}$	-0.040**	-0.040**		
	(0.020)	(0.020)		
$DisMortality_{ic} \times Disaster_{ct-1}$		-0.006		
		(0.004)		
$Leader_{ict-1} \times DisasterLarge_{ct-1}$			-0.027	-0.027
			(0.019)	(0.019)
$DisMortality_{ic} \times DisasterLarge_{ct-1}$				-0.004
				(0.004)
Number of regions	38,427	38,427	38,427	38,427
Observations	690,495	690,495	690,495	690,495
R-squared	0.319	0.319	0.319	0.319
Region FE	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes

Notes: Fixed effect regressions using annual data for subnational regions between 1992 and 2009. $Light_{ict}$ is the logarithm of average nighttime light intensity plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in country c in year t, and 0 otherwise. $Disaster_{ct}$ is a dummy variable equal to 1 if a natural disaster hit country c in year t, and 0 otherwise. $DisasterLarge_{ct}$ is a dummy variable equal to 1 if a natural disaster that affected more than 1,000 people hit country c in year t, and 0 otherwise. $DisMortality_{ic}$ is region i's natural hazard mortality risk. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Table S.11: Leaders' Characteristics and Regional Favoritism

Dependent variable	(1) $Light_{ict}$	(2) $Light_{ict}$	(3) $Light_{ict}$	(4) $Light_{ict}$
$Leader_{ict-1}$	0.060***	0.026	-0.074	0.007
	(0.019)	(0.022)	(0.077)	(0.080)
$Leader_{ict-1} \times Graduate_{ct-1}$	-0.047			-0.020
	(0.035)			(0.037)
$Leader_{ict-1} \times Military_{ct-1}$		0.083		0.116**
		(0.050)		(0.052)
$Leader_{ict-1} \times Lawyer_{ct-1}$		-0.038		-0.042
		(0.039)		(0.043)
$Leader_{ict-1} \times ProfScien_{ct-1}$		-0.037		-0.032
		(0.050)		(0.054)
$Leader_{ict-1} \times Age_{ct-1}$			0.002	0.001
			(0.001)	(0.001)
Number of regions	38,376	38,390	38,390	38,376
Observations	508,466	515,784	534,696	505,377
R-squared	0.291	0.294	0.293	0.291
Region FE	Yes	Yes	Yes	Yes
Country-year FE	Yes	Yes	Yes	Yes

Notes: Fixed effect regressions using annual data for subnational regions between 1992 and 2005. $Light_{ict}$ is the logarithm of average nighttime light intensity plus 0.01. $Leader_{ict}$ is a dummy variable equal to 1 if region i is the birth region of the political leader in t; and 0 otherwise. $Graduate_{ct}$ is a dummy variable equal to 1 if the current political leader has a graduate degree; and 0 otherwise. $Military_{ct}$ is a dummy variable that is equal to 1 if the current political leader's prior occupation was with the military; and 0 otherwise. $Lawyer_{ct}$ is a dummy variable equal to 1 if the current political leader was a lawyer; and 0 otherwise. $ProfScien_{ct}$ is a dummy variable equal to 1 if the current political leader was a professor or scientist; and 0 otherwise. Age_{ct} is the political leader's current age. Standard errors are adjusted for leader clustering. ***, **, * indicate significance at the 1, 5 and 10%-level, respectively.

Figure S.1: Distribution of average nighttime light intensity pixel value, 1992-2009

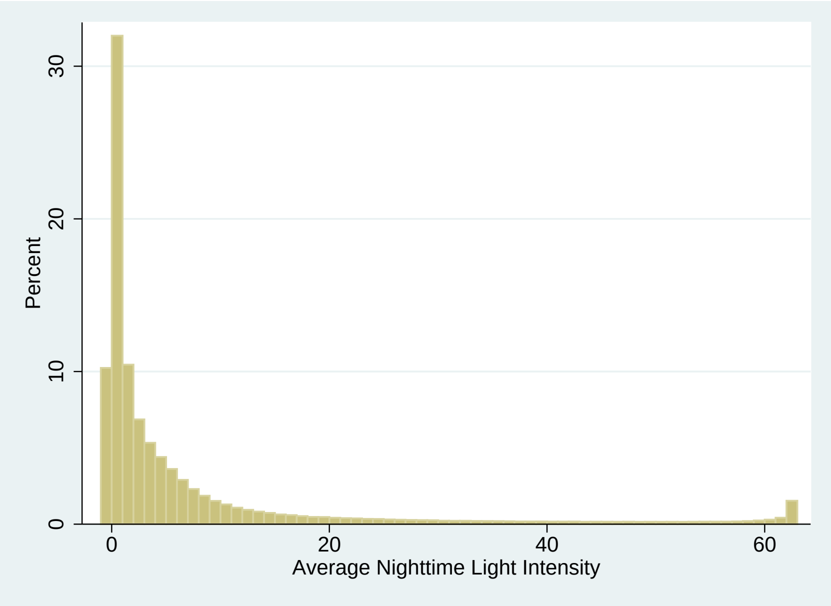


Figure S.2: Distribution of $Light_{ict}$, 1992-2009

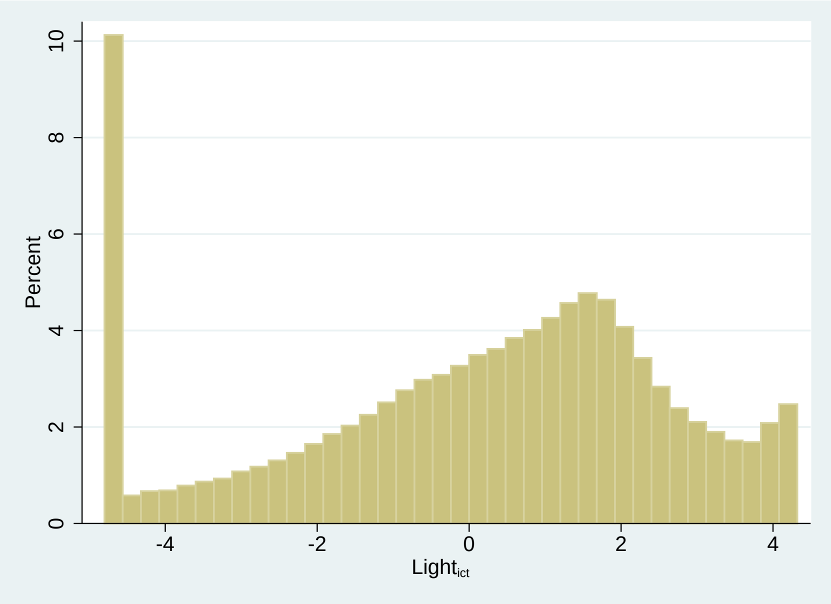
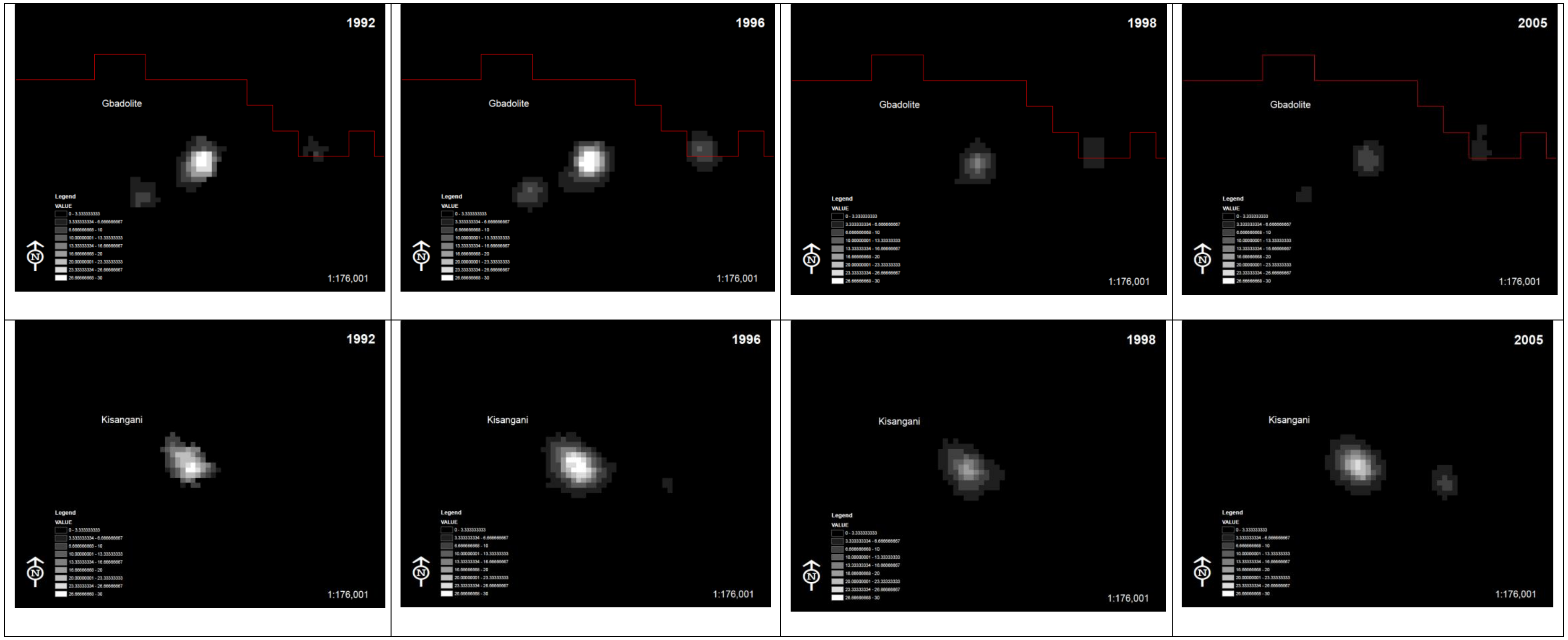
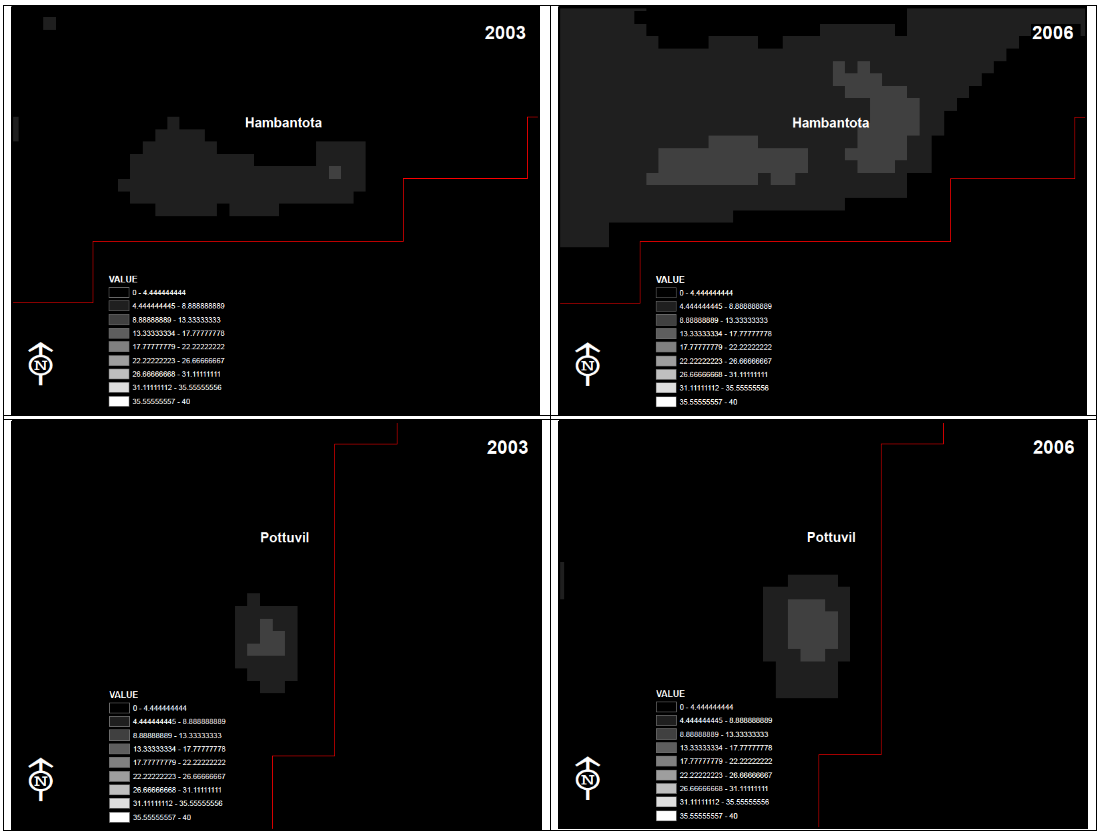


Figure S.3: Nightime light intensity in Gbadolite and Kisangani in 1992, 1996, 1998 and 2005



Notes: The upper panel shows nighttime light intensity in Gbadolite, and the lower panel shows nighttime light intensity in Kisangani. Mobutu Sese Seko was president of Zaire until 1996. The Second Congo War lasted from 1998 to 2003.

Figure S.4: Nightime light intensity in Hambantota and Pottuvil in 2004 and 2006



Notes: The upper panel shows nighttime light intensity in Hambantota, and the lower panel shows nighttime light intensity in Pottuvil. Mahinda Rajapaksa became prime minister of Sri Lanka in 2004, and president in 2005.